Automated Model Quality Estimation and Change Impact Analysis on Model Histories

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ABSTRACT

Cyber-Physical Systems integrate hardware with software in complex applications. To mitigate the complexity, engineers rely on model-based systems engineering approaches. Updates and function enhancements lead to frequently changing design constraints and objectives. These changes increase the need to rework and extend model artifacts of the system. This can cause quality degradation over time due to modeling errors, knowledge disparities, or a lack of guidelines. To enable efficient collaboration and reduce maintenance costs in model-based systems engineering, the industry needs a cost-efficient, scalable approach to monitor model quality. The work outlines a doctoral thesis investigating the potential of automated data-driven quality assessment strategies using model artifact history and model changes. We will extract metrics and model changes to establish quality feedback for system engineers. We aim to use manual model quality assessments to incorporate domain-specific expert knowledge into the automated strategy. The main goals are to lower the effort of model quality assessments, to provide practitioners with foresight on quality development, and to estimate task effort to improve model quality.

KEYWORDS

Model-based Systems Engineering, Model Quality, Model Metrics, Quality Assessment, Model Review, Change-Impact Analysis

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1 INTRODUCTION

Cyber-Physical Systems (CPSs) have a long lifetime and are subject to frequent re-engineering [12, 34]. Many companies use model-based systems engineering (MBSE) to reduce the complexity in CPS development [19, 25, 26]. MBSE is based on abstraction and the single source of truth principle. It offers benefits like code and report generation, verification and validation, improved communication, maintenance, and knowledge distribution [1, 19]. High model quality enables most of these benefits [5]. Model quality targets completeness and correctness, looking from ground level to bird's eye view on model artifacts [32]. The literature proposes many quality models for code quality but less for the model-driven context [15]. Existing examples are SEQUAL [18], QoMo [8], GoM [30] or the 6C [15] framework. Mohaghegi *et al.* identified six major model quality attributes in MBSE: correctness, completeness, consistency, comprehensibility, confinement, and changeability [25].

2 PROBLEM STATEMENT

Metric analysis often lacks an interpretation of the extracted data, while offering a fast result. Metrics alone can not sufficiently describe the subjective nature of quality. Rule-based approaches primarily target syntax and miss a qualitative domain or project-specific evaluation [17, 24, 25]. Manual inspections by domain experts would offer a higher qualitative value [32], but due to the size of industrial CPS models, such assessments are costly and not feasible to be done frequently. Related work additionally motivates to enable frequent, in-depth quality analysis of models:

- Rapidly changing requirements increase re-modeling activities [12], which can disrupt artifact quality, slowing down development and increasing uncertainty [13].
- A lack of operationalization of quality models in MBSE makes it more difficult for the industry to apply them.
- Degradation of quality is connected to the accumulation of technical debt[1] and model debt[2], reducing extensibility while increasing maintenance effort [3, 33, 35].
- A lack of applicable methods and tools that support system engineers in quality assessments of industry-sized models [7, 13, 14, 16, 26, 36].

Automation of a complex artifact or project-specific extensible qualitative assessment is challenging [4, 14, 16, 17]. Missing requirements specification are hindering the possibility of model artifact quality assessments [4, 15]. The lack of monitored metadata of artifacts limits automation further [29]. Related work focused on the definition of quality and quality models [15, 25]. We see a gap in investigating the applicability of quality assessments on modern industrial-sized models. Basciani *et al.* extended tool support in the context of quantitative assessments [7]. Other work focused on versioning of models to first enable investigating model histories [23]. Popoola worked on the classification of design decisions by investigating model changes in model history but without focusing on the impact on model quality [28].

[1]Long-term negative consequences of sub-optimal short-term fixes [10].
[2]Cost of maintaining and updating models over time [20].

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3 RESEARCH HYPOTHESIS

Supported by related work, we assume that in-depth manual assessments can provide a more granular qualitative analysis of model artifacts than strict rule- and syntax-based checks [17, 20, 31]. Our research hypothesis is grounded on the following assertions: **(A1)** The existing literature lacks collected evidence of the applicability, scalability, and inherent limitations of quality models and quality assessment strategies within MBSE; **(A2)** By gathering model metrics and manual quality assessments on model histories, we can accumulate data in a dataset that supports a data-driven quality estimation during modeling to incorporate domain context in the assessment; **(A3)** The historical evolution of metrics and quality attributes provides a basis to predict the impact of model changes on model quality and effort for quality improvement.

To investigate our first assertion (A1), we define the following research questions for a systematic literature review (SLR):

- RQ1 What automated methods currently exist for quality assessment in MBSE?
  - RQ1.1 Which strategies employ metric analysis?
  - RQ1.2 Which methods ground on rule-based approaches?
  - RQ1.3 What other distinctive approaches are there?
- RQ2 What are existing manual methods for assessing the quality of model artifacts in MBSE?
- RQ3 What are the limitations of the collected quality assessment strategies that impact practitioners?

To challenge assertion (A2), we develop a prototype collecting data for quality attribute estimations. The RQs are:

- RQ4 Which quality attributes can we accurately estimate with model artifact metrics from model histories?
- RQ5 What data-driven estimation method performs best with which metrics for which quality attribute?

To challenge assertion (A3), we include model changes in our estimation pipeline. We investigate the impact of changes on quality and experiment with a predictive approach to foresee quality development after changes. The RQs are:

- RQ6 How can we predict the quality evolution of model artifacts with model changes and quality assessment data from model histories?
- RQ7 How can we estimate maintenance effort to improve model quality with model change and quality assessment data from model histories?

4 EXPECTED CONTRIBUTIONS

To improve the automation of quality assessments in MBSE, we will conduct an SLR to collect quality assessment methodologies in MBSE, their degree of automation, and their current limitations. To gather evidence from the state of practice, we plan to incorporate the views of practitioners concerning the results of the SLR in a Rapid Review [9] fashion. This will guide researchers to contribute to the field of applicable quality assessments in MBSE. Our SLR will support the identification of suitable quality attributes applicable to different types of models. Through model history analysis, we will experiment with underrepresented data-driven quality assessments in MBSE due to artifact data scarcity [11, 29]. By incorporating manual assessment data, we plan to introduce domain knowledge and tackle limitations of existing approaches [7, 21]. We hope to enhance the real-time availability of quality assessments and lower the effort for future assessments. For code, an example of continuous quality assessments is SonarQube[3].

We plan to use model histories to investigate maintainability aspects, e.g., model changes, as well as metric evolution, e.g., structural metrics, and quality attribute development over time, to provide a quality estimation, and communicate quality evolution in model artifacts to practitioners. The biggest limitation we see arises from the availability and quality of data.

5 METHODOLOGY

The methodology section follows the research questions.

The SLR [22] ensures the comprehensiveness and repeatability of the findings targeting [RQ1].

For [RQ2], we plan to collect model history data within the MBSE practical course offered at fortiss in collaboration with TU Munich. Two teams work in parallel on comprehensive models of autonomous driving functions using the open-source tool AutoFOCUS3[4] (AF3) that are deployed onto the fortissimo[5] miniature cars. AF3 enables graphical component-based modeling of systems based on the SPES [27] methodology in terms of its different viewpoints [2, 6], e.g., requirements, logical architecture (including behavior), technical HW and SW architecture, deployment.

We will manually assess the quality of our models according to suitable assessment strategies collected in the SLR. We will document the effort for the manual assessments. To look at the potential savings in effort, we experiment with our data-driven approach to estimate the quality of different model versions in a test set. We will further compare and investigate the accuracy of our prototype.

To tackle data scarcity and increase industrial relevancy, we are preparing to conduct case studies with industry partners from the fortiss network, where we plan to provide them with a prototype implementation of our approach. For industry-sized models, we will need to adjust the underlying quality assessment according to the available project histories. A case study adds insights into the current industry needs that are not prominent in the literature. Further, surveys can query the perceived benefits for practitioners.

Though open-source system models are rare, we additionally investigate existing models for our use case to find evidence for the applicability of our approach. Examples are ModelSet[6] or the TMT SysML model[7], where we will search for relevant model histories to be annotated.

For [RQ3], we continue with the integration of the analysis of model changes. Model changes can be extracted in our controlled environment of AF3, by tracking them while modeling. In industrial models, the extraction of changes can rely on a comparison of model versions to obtain diffs. We can, therefore, use the development of metrics, the development of the models, and the development of the quality attributes as input data. We continue to estimate either what impact different changes have on model quality or what changes might be necessary to impact model quality positively.

[3]https://www.sonarsource.com/products/sonarqube/
[4]https://af3.fortiss.org/
[5]https://www.fortiss.org/en/research/fortiss-labs/detail/mobility-lab/
[6]https://modelset.github.io/
[7]https://github.com/Open-MBEE/TMT-SysML-Model/

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